Classical Restoration of Blurred License Plate Images

Does a Higher PSNR Mean the Plate Is Easier to Read?

Project Proposal, CS7.404 Digital Image Processing, Monsoon 2026

Instructor: Prof. Raghvendra G S

Team: Ishit Agarwal (2026909005), Ashish Beniwal (2026909001), S. Ujwal (2026701052)

---

1. Problem Statement

License plates captured by traffic and CCTV cameras are often unreadable because of motion blur, defocus, low resolution and sensor noise. We model the observed plate as

$$g(x, y) = \mathcal{D}\big[\, h(x, y) * f(x, y) \,\big] + n(x, y), \tag{1}$$

where f is the sharp plate, h is the blur kernel (point spread function, PSF), $\mathcal{D}$ is downsampling and n is additive noise. The goal is to recover f from g using classical restoration methods implemented from scratch.

Restoration is usually judged with PSNR and SSIM, which measure how close the output is to the original image. For plates, what matters is whether the characters can be read. These two goals may not agree: a smoother image can score a higher PSNR while losing the thin strokes that separate an `8` from a `B`. This project measures that gap, using OCR as an objective test of legibility.

2. Research Questions

Q1. (Primary) Does the restoration setting with the best PSNR also give the best OCR accuracy? We sweep each filter's parameter and compare the PSNR-optimal and accuracy-optimal settings across blur and noise levels.

Q2. How much accuracy is lost when the blur must be estimated from the image (*blind*) instead of being known (*oracle*)?

Q3. Does restoration increase confident misreads? Every OCR result is labelled as a *correct read*, an *honest failure* (wrong, low confidence) or a *silent failure* (wrong, high confidence).

3. Workflow

Each plate passes through the eight stages in Figure 1. The clean image and its true text are kept alongside, so every result can be scored against ground truth.

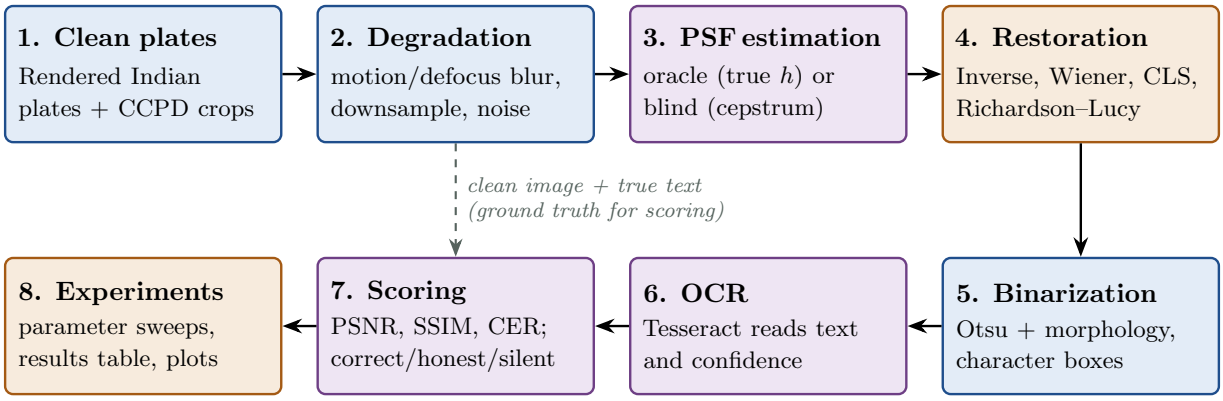


Figure 1: Project workflow.

4. Dataset

- **Synthetic plates (∼600):** random valid Indian registration numbers (e.g. `KA 05 MN 1234`) rendered in a plate-style font. Exact ground truth and no privacy concerns.
- **CCPD base subset (∼400)**: real Chinese plates, cropped and rectified using the corner points stored in each filename. The first (Chinese) character is excluded from scoring.
- **CCPD blur subset (∼50):** real blurred plates, used for a qualitative before/after gallery since they have no clean reference.

Degradation grid: motion blur length 0–21 px at random angles, defocus radius 0–5 px, noise at $40, 30, 25, 20$ dB BSNR, and downsampling by a factor of 1 or 2.

5. Methods and Implementation

All image processing algorithms are written by the team. Libraries are used only for primitives: NumPy (arrays, FFT), OpenCV (image I/O, rotation, connected components), PIL (text rendering), Tesseract (OCR, as the measurement tool) and Matplotlib/Pandas (plots and tables).

Stage	Implemented from scratch
Degradation	Motion and defocus PSF generation, FFT-based blurring, block-mean downsampling, noise at a target BSNR
Blind PSF estimation	Blur angle from the orientation of spectral stripes; blur length from the negative peak of the cepstrum
Restoration	Inverse filter, Wiener filter, constrained least squares, Richardson–Lucy, edge tapering
Binarization	Otsu thresholding, erosion, dilation, opening, closing, character filtering
Evaluation	PSNR, SSIM, character error rate (edit distance), outcome classifier, experiment runner

Restoration filters (with G, H the Fourier transforms of g, h, and P the Laplacian):

$$\text{Wiener:} \quad \hat{F}(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + K}\, G(u,v)$$

$$\text{CLS:} \quad \hat{F}(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + \gamma\,|P(u,v)|^2}\, G(u,v)$$

$$\text{Richardson–Lucy:} \quad \hat{f}_{k+1} = \hat{f}_k \cdot \left[\tilde{h} * \frac{g}{h * \hat{f}_k}\right], \qquad \tilde{h}(x,y) = h(-x,-y)$$

6. Evaluation

Restored images are scored for fidelity (PSNR, SSIM) and for legibility (character accuracy from Tesseract, via edit distance against the true text). Each OCR result is also classified as a correct read, an honest failure or a silent failure using a confidence threshold fixed in advance. We sweep each filter's parameter (Wiener K, CLS γ, Richardson–Lucy iterations) across all blur and noise levels, with both the true and the estimated PSF. This lets us compare PSNR-optimal and accuracy-optimal settings (Q1), measure the cost of blind estimation (Q2), track confident misreads (Q3), and map the blur and noise levels beyond which plates can no longer be read. Real blurred CCPD plates are used for a qualitative before/after check.